

# Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

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Answer Part A by circling one option. Answer Part B in full, showing your working. A periodic table and the answer key are at the back.

## Part A · Multiple choice

A1 Topic: Energy & Matters EM-LO1

Which statement best describes **heat**?

- A. The total energy stored in an object
- B. The transfer of energy between objects because of a temperature difference
- C. The temperature of an object
- D. The energy stored in chemical bonds

A2 Topic: Energy & Matters EM-LO1

A snack provides 250 Cal (nutritional Calories). What is this in kJ? (1 Cal = 1 kcal = 4.184 kJ)

- A. 59.8 kJ
- B. 250 kJ
- C.  $1.05 \times 10^3$  kJ
- D.  $1.05 \times 10^6$  kJ

**A3** Topic: Energy & Matters EM-LO2

Normal body temperature is 37.0 °C. What is this in kelvin?

- A. 236.2 K
- B. 98.6 K
- C. 310.2 K
- D. 273.2 K

**A4** Topic: Energy & Matters EM-LO2

The same amount of heat is added to 10 g samples of water, ethanol, aluminium and copper, all starting at 25 °C. Which shows the largest temperature rise?

- A. water (4.184 J/g·°C)
- B. ethanol (2.418 J/g·°C)
- C. aluminium (0.900 J/g·°C)
- D. copper (0.385 J/g·°C)

**A5** Topic: Energy & Matters EM-LO3

Which phase change is **exothermic**?

- A. Melting
- B. Boiling
- C. Sublimation
- D. Freezing

**A6** Topic: Energy & Matters EM-LO3

How much heat is needed to vaporise 10.0 g of water at 100 °C? ( $\Delta H_{\text{vap}} = 2260 \text{ J/g}$ )

- A. 3.34 kJ
- B. 2.26 kJ
- C. 22.6 kJ
- D. 41.8 kJ

**A7** Topic: Energy & Matters EM-LO4

On the heating curve of water, what is happening along a flat (horizontal) segment?

- A. The temperature rises steadily
- B. Heat is added but the temperature stays constant while the phase changes
- C. No heat is being added
- D. The substance is cooling down

**A8** Topic: Energy & Matters EM-LO4

Which segment of water's heating curve is calculated with  $q = m \times \Delta H_{\text{vap}}$ ?

- A. The rising line below 0 °C
- B. The plateau at 0 °C
- C. The plateau at 100 °C
- D. The rising line above 100 °C

**A9** Topic: Energy & Matters EM-LO5

In a reaction, breaking the bonds in the reactants needs 2000 kJ and forming the bonds in the products releases 2400 kJ. What is  $\Delta H$ ?

- A. +400 kJ, endothermic
- B. -400 kJ, exothermic
- C. +4400 kJ, endothermic
- D. -4400 kJ, exothermic

**A10** Topic: Energy & Matters EM-LO6

Glucose is oxidised in the body:  $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$ , releasing about 670 kcal/mol. Which statement is correct?

- A. The reaction is endothermic and  $\Delta H$  is positive
- B. The reaction is exothermic and  $\Delta H$  is negative
- C. The reaction absorbs energy from ATP
- D. No bonds are broken in the reaction

## Part B · Structured questions

Show all working. The marks and Bloom's level are shown for each question.

### Question B1 Topic: Energy & Matters EM-LO1 5 marks, C3

(a) Define **energy** and **heat**. (2 marks) (b) An oral nutrition snack bar contains 8.0 g of fat, 30. g of carbohydrate and 5.0 g of protein. Using the average fuel values (fat 9 Cal/g; carbohydrate and protein 4 Cal/g), calculate its energy content in Calories and in kJ. (3 marks)

### Question B2 Topic: Energy & Matters EM-LO2 6 marks, C3

(a) A patient's temperature is 101.3 °F. Convert it to °C and to K, and state whether the patient has a fever. (2 marks) (b) Calculate the heat, in kJ, needed to warm 250. g of water from 22.0 °C to 85.0 °C ( $c = 4.184 \text{ J/g}\cdot^\circ\text{C}$ ). (2 marks) (c) A 40.0 g metal sample absorbs 900. J and its temperature rises from 18.0 °C to 43.0 °C. Calculate its specific heat and identify the metal using the specific heat table in the slides. (2 marks)

**Question B3** Topic: Energy & Matters EM-LO3 6 marks, C3

A 25.0 g ice pack is taken from a freezer at  $-10.0\text{ }^{\circ}\text{C}$  and left until it becomes liquid water at  $25.0\text{ }^{\circ}\text{C}$ . (a) Name the three stages of the heating curve involved. (1 mark) (b) Calculate the total heat absorbed, in kJ. ( $c$  of ice =  $2.06\text{ J/g}\cdot^{\circ}\text{C}$ ;  $\Delta H_{\text{fus}} = 334\text{ J/g}$ ;  $c$  of water =  $4.184\text{ J/g}\cdot^{\circ}\text{C}$ ) (4 marks) (c) Explain why the temperature stays at  $0\text{ }^{\circ}\text{C}$  while the ice melts. (1 mark)

**Question B4** Topic: Energy & Matters EM-LO5 5 marks, C3

(a) Define **bond energy**. (1 mark) (b) Use the bond energies below to estimate  $\Delta H$  for:  $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$ . Bond energies (kJ/mol): H-H 439, Cl-Cl 243, H-Cl 431. (3 marks) (c) State whether the reaction is endothermic or exothermic, and whether heat is written on the reactant or product side of the equation. (1 mark)

**Question B5** Topic: Energy & Matters EM-LO6 5 marks, C3

(a) A patient receives 25.0 g of glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ , 180.2 g/mol). Calculate the energy released when it is fully oxidised, in kJ and in kcal. (Glucose releases 2800 kJ/mol, or 670 kcal/mol.) (3 marks) (b) Describe what happens to the bonds when ATP is converted to ADP, and state the approximate energy change. (2 marks)

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**Periodic table** · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

| 1A<br>1                          | 2A<br>2                          | 3                               | 4                                | 5                                | 6                               | 7                                | 8                                | 9                                | 10                               | 11                               | 12                               | 3A<br>13                         | 4A<br>14                         | 5A<br>15                         | 6A<br>16                         | 7A<br>17                        | 8A<br>18                       |
|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|---------------------------------|--------------------------------|
| 1<br><b>H</b><br>1.01<br>2.1     |                                  |                                 |                                  |                                  |                                 |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                  |                                 | 2<br><b>He</b><br>4.00<br>—    |
| 3<br><b>Li</b><br>6.94<br>1.0    | 4<br><b>Be</b><br>9.01<br>1.5    |                                 |                                  |                                  |                                 |                                  |                                  |                                  |                                  |                                  |                                  | 5<br><b>B</b><br>10.81<br>2.0    | 6<br><b>C</b><br>12.01<br>2.5    | 7<br><b>N</b><br>14.01<br>3.0    | 8<br><b>O</b><br>16.00<br>3.5    | 9<br><b>F</b><br>19.00<br>4.0   | 10<br><b>Ne</b><br>20.18<br>—  |
| 11<br><b>Na</b><br>22.99<br>0.9  | 12<br><b>Mg</b><br>24.31<br>1.2  |                                 |                                  |                                  |                                 |                                  |                                  |                                  |                                  |                                  |                                  | 13<br><b>Al</b><br>26.98<br>1.5  | 14<br><b>Si</b><br>28.09<br>1.8  | 15<br><b>P</b><br>30.97<br>2.1   | 16<br><b>S</b><br>32.07<br>2.5   | 17<br><b>Cl</b><br>35.45<br>3.0 | 18<br><b>Ar</b><br>39.95<br>—  |
| 19<br><b>K</b><br>39.10<br>0.8   | 20<br><b>Ca</b><br>40.08<br>1.0  | 21<br><b>Sc</b><br>44.96<br>1.3 | 22<br><b>Ti</b><br>47.87<br>1.5  | 23<br><b>V</b><br>50.94<br>1.6   | 24<br><b>Cr</b><br>52.00<br>1.6 | 25<br><b>Mn</b><br>54.94<br>1.5  | 26<br><b>Fe</b><br>55.84<br>1.8  | 27<br><b>Co</b><br>58.93<br>1.9  | 28<br><b>Ni</b><br>58.69<br>1.9  | 29<br><b>Cu</b><br>63.55<br>1.9  | 30<br><b>Zn</b><br>65.38<br>1.6  | 31<br><b>Ga</b><br>69.72<br>1.6  | 32<br><b>Ge</b><br>72.63<br>1.8  | 33<br><b>As</b><br>74.92<br>2.0  | 34<br><b>Se</b><br>78.97<br>2.4  | 35<br><b>Br</b><br>79.90<br>2.8 | 36<br><b>Kr</b><br>83.80<br>—  |
| 37<br><b>Rb</b><br>85.47<br>0.8  | 38<br><b>Sr</b><br>87.62<br>1.0  | 39<br><b>Y</b><br>88.91<br>1.2  | 40<br><b>Zr</b><br>91.22<br>1.4  | 41<br><b>Nb</b><br>92.91<br>1.6  | 42<br><b>Mo</b><br>95.95<br>1.8 | 43<br><b>Tc</b><br>[98]<br>1.9   | 44<br><b>Ru</b><br>101.07<br>2.2 | 45<br><b>Rh</b><br>102.91<br>2.2 | 46<br><b>Pd</b><br>106.42<br>2.2 | 47<br><b>Ag</b><br>107.87<br>1.9 | 48<br><b>Cd</b><br>112.41<br>1.7 | 49<br><b>In</b><br>114.82<br>1.7 | 50<br><b>Sn</b><br>118.71<br>1.8 | 51<br><b>Sb</b><br>121.76<br>1.9 | 52<br><b>Te</b><br>127.60<br>2.1 | 53<br><b>I</b><br>126.90<br>2.5 | 54<br><b>Xe</b><br>131.29<br>— |
| 55<br><b>Cs</b><br>132.91<br>0.7 | 56<br><b>Ba</b><br>137.33<br>0.9 | 57–71                           | 72<br><b>Hf</b><br>178.49<br>1.3 | 73<br><b>Ta</b><br>180.95<br>1.5 | 74<br><b>W</b><br>183.84<br>1.7 | 75<br><b>Re</b><br>186.21<br>1.9 | 76<br><b>Os</b><br>190.23<br>2.2 | 77<br><b>Ir</b><br>192.22<br>2.2 | 78<br><b>Pt</b><br>195.08<br>2.2 | 79<br><b>Au</b><br>196.97<br>2.4 | 80<br><b>Hg</b><br>200.59<br>1.9 | 81<br><b>Tl</b><br>204.38<br>1.8 | 82<br><b>Pb</b><br>207.2<br>1.9  | 83<br><b>Bi</b><br>208.98<br>1.9 | 84<br><b>Po</b><br>[209]<br>2.0  | 85<br><b>At</b><br>[210]<br>2.2 | 86<br><b>Rn</b><br>[222]<br>—  |
| 87<br><b>Fr</b><br>[223]<br>0.7  | 88<br><b>Ra</b><br>[226]<br>0.9  | 89–103                          | 104<br><b>Rf</b><br>[267]<br>—   | 105<br><b>Db</b><br>[268]<br>—   | 106<br><b>Sg</b><br>[269]<br>—  | 107<br><b>Bh</b><br>[270]<br>—   | 108<br><b>Hs</b><br>[277]<br>—   | 109<br><b>Mt</b><br>[278]<br>—   | 110<br><b>Ds</b><br>[281]<br>—   | 111<br><b>Rg</b><br>[282]<br>—   | 112<br><b>Cn</b><br>[285]<br>—   | 113<br><b>Nh</b><br>[286]<br>—   | 114<br><b>Fl</b><br>[289]<br>—   | 115<br><b>Mc</b><br>[290]<br>—   | 116<br><b>Lv</b><br>[293]<br>—   | 117<br><b>Ts</b><br>[294]<br>—  | 118<br><b>Og</b><br>[294]<br>— |
| Lanthanides                      |                                  | 57<br><b>La</b><br>138.91<br>—  | 58<br><b>Ce</b><br>140.12<br>—   | 59<br><b>Pr</b><br>140.91<br>—   | 60<br><b>Nd</b><br>144.24<br>—  | 61<br><b>Pm</b><br>[145]<br>—    | 62<br><b>Sm</b><br>150.36<br>—   | 63<br><b>Eu</b><br>151.96<br>—   | 64<br><b>Gd</b><br>157.25<br>—   | 65<br><b>Tb</b><br>158.93<br>—   | 66<br><b>Dy</b><br>162.50<br>—   | 67<br><b>Ho</b><br>164.93<br>—   | 68<br><b>Er</b><br>167.26<br>—   | 69<br><b>Tm</b><br>168.93<br>—   | 70<br><b>Yb</b><br>173.05<br>—   | 71<br><b>Lu</b><br>174.97<br>—  |                                |
| Actinides                        |                                  | 89<br><b>Ac</b><br>[227]<br>1.1 | 90<br><b>Th</b><br>232.04<br>1.3 | 91<br><b>Pa</b><br>231.04<br>1.4 | 92<br><b>U</b><br>238.03<br>1.4 | 93<br><b>Np</b><br>[237]<br>—    | 94<br><b>Pu</b><br>[244]<br>—    | 95<br><b>Am</b><br>[243]<br>—    | 96<br><b>Cm</b><br>[247]<br>—    | 97<br><b>Bk</b><br>[247]<br>—    | 98<br><b>Cf</b><br>[251]<br>—    | 99<br><b>Es</b><br>[252]<br>—    | 100<br><b>Fm</b><br>[257]<br>—   | 101<br><b>Md</b><br>[258]<br>—   | 102<br><b>No</b><br>[259]<br>—   | 103<br><b>Lr</b><br>[262]<br>—  |                                |

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

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## Answer key

### Part A · Multiple choice

A1 Topic: Energy & Matters EM-LO1

Which statement best describes **heat**?

- A. The total energy stored in an object
- B. The transfer of energy between objects because of a temperature difference**
- C. The temperature of an object
- D. The energy stored in chemical bonds

Heat is energy in transit: it flows from the hotter object to the colder one. An object contains energy; heat is what flows.

A2 Topic: Energy & Matters EM-LO1

A snack provides 250 Cal (nutritional Calories). What is this in kJ? (1 Cal = 1 kcal = 4.184 kJ)

- A. 59.8 kJ
- B. 250 kJ
- C.  $1.05 \times 10^3$  kJ**
- D.  $1.05 \times 10^6$  kJ

$250 \text{ kcal} \times 4.184 \text{ kJ/kcal} = 1046 \text{ kJ} = 1.05 \times 10^3 \text{ kJ}$ . ( $1.05 \times 10^6$  is the value in joules.)

**A3** Topic: Energy & Matters EM-LO2

Normal body temperature is 37.0 °C. What is this in kelvin?

- A. 236.2 K
- B. 98.6 K
- C. 310.2 K**
- D. 273.2 K

$$K = ^\circ C + 273.15 = 37.0 + 273.15 = 310.2 \text{ K.}$$

**A4** Topic: Energy & Matters EM-LO2

The same amount of heat is added to 10 g samples of water, ethanol, aluminium and copper, all starting at 25 °C. Which shows the largest temperature rise?

- A. water (4.184 J/g·°C)
- B. ethanol (2.418 J/g·°C)
- C. aluminium (0.900 J/g·°C)
- D. copper (0.385 J/g·°C)**

$\Delta T = q \div (m \times c)$ . With the same  $q$  and  $m$ , the smallest specific heat gives the largest  $\Delta T$ .

**A5** Topic: Energy & Matters EM-LO3

Which phase change is **exothermic**?

- A. Melting
- B. Boiling
- C. Sublimation
- D. Freezing**

Freezing removes heat from the substance, so it releases heat (exothermic). Melting, boiling and sublimation all absorb heat.

**A6** Topic: Energy & Matters EM-LO3

How much heat is needed to vaporise 10.0 g of water at 100 °C? ( $\Delta H_{\text{vap}} = 2260 \text{ J/g}$ )

- A. 3.34 kJ
- B. 2.26 kJ
- C. 22.6 kJ**
- D. 41.8 kJ

$$q = m \times \Delta H_{\text{vap}} = 10.0 \text{ g} \times 2260 \text{ J/g} = 22\,600 \text{ J} = 22.6 \text{ kJ}.$$

**A7** Topic: Energy & Matters EM-LO4

On the heating curve of water, what is happening along a flat (horizontal) segment?

- A. The temperature rises steadily
- B. Heat is added but the temperature stays constant while the phase changes**
- C. No heat is being added
- D. The substance is cooling down

On a plateau the added heat is used to change the phase (melting or boiling), not to raise the temperature.

**A8** Topic: Energy & Matters EM-LO4

Which segment of water's heating curve is calculated with  $q = m \times \Delta H_{\text{vap}}$ ?

- A. The rising line below 0 °C
- B. The plateau at 0 °C
- C. The plateau at 100 °C**
- D. The rising line above 100 °C

The plateau at 100 °C is boiling (liquid → gas), which uses the heat of vaporisation. The plateau at 0 °C uses  $\Delta H_{\text{fus}}$ ; the rising lines use  $q = mc\Delta T$ .

**A9** Topic: Energy & Matters EM-LO5

In a reaction, breaking the bonds in the reactants needs 2000 kJ and forming the bonds in the products releases 2400 kJ. What is  $\Delta H$ ?

- A. +400 kJ, endothermic
- B. -400 kJ, exothermic**
- C. +4400 kJ, endothermic
- D. -4400 kJ, exothermic

$\Delta H \approx \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed}) = 2000 - 2400 = -400 \text{ kJ}$ . Negative, so exothermic.

**A10** Topic: Energy & Matters EM-LO6

Glucose is oxidised in the body:  $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$ , releasing about 670 kcal/mol. Which statement is correct?

- A. The reaction is endothermic and  $\Delta H$  is positive
- B. The reaction is exothermic and  $\Delta H$  is negative**
- C. The reaction absorbs energy from ATP
- D. No bonds are broken in the reaction

Energy is released, so the reaction is exothermic and  $\Delta H$  is negative. The bonds formed in  $\text{CO}_2$  and  $\text{H}_2\text{O}$  are stronger overall than the bonds broken.

## Part B · Structured questions

Question B1   Topic: Energy & Matters   EM-LO1   5 marks, C3

(a) Define **energy** and **heat**. (2 marks) (b) An oral nutrition snack bar contains 8.0 g of fat, 30. g of carbohydrate and 5.0 g of protein. Using the average fuel values (fat 9 Cal/g; carbohydrate and protein 4 Cal/g), calculate its energy content in Calories and in kJ. (3 marks)

(a) **Energy**: the ability to do work. **Heat**: the transfer of energy between objects because they have different temperatures. (b) Fat:  $8.0 \text{ g} \times 9 \text{ Cal/g} = 72 \text{ Cal}$ . Carbohydrate + protein:  $(30. + 5.0) \text{ g} \times 4 \text{ Cal/g} = 140 \text{ Cal}$ . Total = **212 Cal**.  $212 \text{ Cal} \times 4.184 \text{ kJ/Cal} = \mathbf{887 \text{ kJ}}$ .

| Step | Marking criteria                                             | Marks    |
|------|--------------------------------------------------------------|----------|
| 1    | Energy: the ability to do work                               | 1        |
| 2    | Heat: energy transferred because of a temperature difference | 1        |
| 3    | Fat: 72 Cal                                                  | 1        |
| 4    | Carbohydrate + protein: 140 Cal                              | 1        |
| 5    | 212 Cal and 887 kJ                                           | 1        |
|      | <b>Total</b>                                                 | <b>5</b> |

**Question B2**

Topic: Energy &amp; Matters

EM-LO2

6 marks, C3

(a) A patient's temperature is 101.3 °F. Convert it to °C and to K, and state whether the patient has a fever. (2 marks) (b) Calculate the heat, in kJ, needed to warm 250. g of water from 22.0 °C to 85.0 °C ( $c = 4.184 \text{ J/g}\cdot^\circ\text{C}$ ). (2 marks) (c) A 40.0 g metal sample absorbs 900. J and its temperature rises from 18.0 °C to 43.0 °C. Calculate its specific heat and identify the metal using the specific heat table in the slides. (2 marks)

(a)  $^\circ\text{C} = (101.3 - 32) \times 5/9 = \mathbf{38.5\ ^\circ\text{C}}$ ;  $\text{K} = 38.5 + 273.15 = \mathbf{311.7\ \text{K}}$ . Above 37.8 °C, so **yes, a fever**. (b)  $\Delta T = 85.0 - 22.0 = 63.0\ ^\circ\text{C}$ ;  $q = 250. \text{ g} \times 4.184 \text{ J/g}\cdot^\circ\text{C} \times 63.0\ ^\circ\text{C} = 65\,898 \text{ J} = \mathbf{65.9\ \text{kJ}}$ . (c)  $\Delta T = 43.0 - 18.0 = 25.0\ ^\circ\text{C}$ ;  $c = 900. \text{ J} \div (40.0 \text{ g} \times 25.0\ ^\circ\text{C}) = \mathbf{0.900\ \text{J/g}\cdot^\circ\text{C}}$ , which matches **aluminium**.

| Step | Marking criteria                                                       | Marks    |
|------|------------------------------------------------------------------------|----------|
| 1    | 38.5 °C and fever stated                                               | 1        |
| 2    | 311.7 K                                                                | 1        |
| 3    | $\Delta T = 63.0\ ^\circ\text{C}$ with correct $q = mc\Delta T$ set-up | 1        |
| 4    | 65.9 kJ                                                                | 1        |
| 5    | $c = 0.900 \text{ J/g}\cdot^\circ\text{C}$                             | 1        |
| 6    | Aluminium                                                              | 1        |
|      | <b>Total</b>                                                           | <b>6</b> |

**Question B3** Topic: Energy & Matters EM-LO3 6 marks, C3

A 25.0 g ice pack is taken from a freezer at  $-10.0\text{ }^{\circ}\text{C}$  and left until it becomes liquid water at  $25.0\text{ }^{\circ}\text{C}$ . (a) Name the three stages of the heating curve involved. (1 mark) (b) Calculate the total heat absorbed, in kJ. (c of ice =  $2.06\text{ J/g}\cdot^{\circ}\text{C}$ ;  $\Delta H_{\text{fus}} = 334\text{ J/g}$ ; c of water =  $4.184\text{ J/g}\cdot^{\circ}\text{C}$ ) (4 marks) (c) Explain why the temperature stays at  $0\text{ }^{\circ}\text{C}$  while the ice melts. (1 mark)

(a) Ice warms from  $-10.0\text{ }^{\circ}\text{C}$  to  $0\text{ }^{\circ}\text{C}$  → ice melts at  $0\text{ }^{\circ}\text{C}$  → water warms from  $0\text{ }^{\circ}\text{C}$  to  $25.0\text{ }^{\circ}\text{C}$ . (b)  $q_1 = 25.0 \times 2.06 \times 10.0 = 515\text{ J}$ .  $q_2 = 25.0 \times 334 = 8350\text{ J}$ .  $q_3 = 25.0 \times 4.184 \times 25.0 = 2615\text{ J}$ . Total =  $11\,480\text{ J} = \mathbf{11.5\text{ kJ}}$ . (c) The heat is used to overcome the forces holding the molecules in the solid (to change the phase), not to raise the temperature.

| Step | Marking criteria                                         | Marks    |
|------|----------------------------------------------------------|----------|
| 1    | Three stages named in order                              | 1        |
| 2    | $q_1 = 515\text{ J}$                                     | 1        |
| 3    | $q_2 = 8350\text{ J}$                                    | 1        |
| 4    | $q_3 = 2615\text{ J}$                                    | 1        |
| 5    | Total $11.5\text{ kJ}$                                   | 1        |
| 6    | Heat used to change the phase, not raise the temperature | 1        |
|      | <b>Total</b>                                             | <b>6</b> |

**Question B4** Topic: Energy & Matters EM-LO5 5 marks, C3

(a) Define **bond energy**. (1 mark) (b) Use the bond energies below to estimate  $\Delta H$  for:  $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$ . Bond energies (kJ/mol): H-H 439, Cl-Cl 243, H-Cl 431. (3 marks) (c) State whether the reaction is endothermic or exothermic, and whether heat is written on the reactant or product side of the equation. (1 mark)

(a) Bond energy: the energy needed to break one mole of a particular bond (the same amount is released when that bond forms). (b) Broken:  $\text{H-H} + \text{Cl-Cl} = 439 + 243 = 682 \text{ kJ}$ . Formed:  $2 \text{ H-Cl} = 2 \times 431 = 862 \text{ kJ}$ .  $\Delta H \approx 682 - 862 = -180 \text{ kJ}$ . (c)  $\Delta H$  is negative, so **exothermic**; heat is written on the **product side**.

| Step | Marking criteria                     | Marks    |
|------|--------------------------------------|----------|
| 1    | Bond energy defined                  | 1        |
| 2    | Bonds broken: 682 kJ                 | 1        |
| 3    | Bonds formed: 862 kJ                 | 1        |
| 4    | $\Delta H \approx -180 \text{ kJ}$   | 1        |
| 5    | Exothermic, heat on the product side | 1        |
|      | <b>Total</b>                         | <b>5</b> |

**Question B5** Topic: Energy & Matters EM-LO6 5 marks, C3

(a) A patient receives 25.0 g of glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ , 180.2 g/mol). Calculate the energy released when it is fully oxidised, in kJ and in kcal. (Glucose releases 2800 kJ/mol, or 670 kcal/mol.) (3 marks) (b) Describe what happens to the bonds when ATP is converted to ADP, and state the approximate energy change. (2 marks)

(a) Moles =  $25.0 \text{ g} \div 180.2 \text{ g/mol} = 0.139 \text{ mol}$ . Energy =  $0.139 \text{ mol} \times 2800 \text{ kJ/mol} = \mathbf{388 \text{ kJ}}$ ;  $0.139 \text{ mol} \times 670 \text{ kcal/mol} = \mathbf{93.0 \text{ kcal}}$ . (b) An O-P bond breaks, and an O-P bond and two O-H bonds form. Overall about **7.5 kcal/mol (31 kJ/mol) is released**, which the cell uses to drive other reactions.

| Step | Marking criteria                        | Marks    |
|------|-----------------------------------------|----------|
| 1    | 0.139 mol glucose                       | 1        |
| 2    | 388 kJ                                  | 1        |
| 3    | 93.0 kcal                               | 1        |
| 4    | O-P broken; O-P and two O-H formed      | 1        |
| 5    | About 7.5 kcal/mol (31 kJ/mol) released | 1        |
|      | <b>Total</b>                            | <b>5</b> |

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